Nazmus Sakib

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Google Scholar | Portfolio | GitHub | LinkedIn

Research Interests: AI for Software Engineering and SRE, LLM Post-Training, AI Safety, NLP, Agentic Systems

Education →

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering; CGPA: 3.82/4.00 — *Jan 2020 – Mar 2025*

- Academic Rank: Top 20% of class based on academic performance.
- Awards: Enlisted thrice in the Dean's List for academic excellence.

Academic Experience

BRAC University — Dhaka, Bangladesh
Lecturer — *Present*

- Instructing undergraduate courses including **Machine Learning**, **Operating Systems**, **Software Engineering**, and **System Analysis & Design**.
- Mentoring students on architectural patterns and bridging the gap between theoretical concepts and industry application.

UIUC++ Summer Research Internship in Software Engineering (SRSE) — Remote
Mentors: Prof. Darko Marinov (UIUC), Prof. Tianyin Xu (UIUC), Prof. Wenxi Wang (UVA) — *Summer 2026*

- Ongoing work on two projects: a **Mutant-Study** project and a survey with accompanying experiments.
- Turning Failure Postmortems into SRE-Benchmark Problems [Blog]

Publications

Peer-Reviewed

Passing Without Repairing: Verifier Blind Spots in a Live SRE Agent Benchmark
Nazmus Sakib. *NeurIPS 2026 Workshop: Who Verifies the Agents? Toward Reliable Agent Development*, Sydney, Australia, 2026. [Paper] [Code]

MemAgent: A Cache-Inspired Framework for Augmenting Conversational Web Agents With Task-Specific Information
Nazmus Sakib, Protoy Barai, Sifat Ishmam Parisa, Anindya Iqbal. *WOA 2025: 26th Workshop "From Objects to Agents"*, CEUR Workshop Proceedings, Vol. 4028, pp. 106–123, 2025. [Paper] [Code]

Preprints & Under Review

Seen but Not Bound: LLM Operations Agents Treat Confirmation Policies as a Caution Dial, Not a Gate
Nazmus Sakib. *Under review*, 2026. [Code]

Luck Is Not Skill: When Do Paired Rollouts Help Group-Relative RL of LLM Agents?
Nazmus Sakib. *arXiv preprint arXiv:2609.24144*, 2026. [arXiv] [Code]

Industry Experience

Sluqe — Warszawa, Poland (Remote)
Software Engineer (Full Stack & AI) — *May 2025 – May 2026*

- **Sluqe Platform:** Built an AI voice-note platform on **Next.js** and **Firebase Cloud Functions** where recordings are transcribed, structured, and vector-indexed into a searchable corpus; shipped **Genkit** agent flows with in-chat web search and deep research over **Groq**, a **Flutter** iOS/Android client, and **Stripe** tiered billing with server-enforced usage quotas.
- **SOS Alert System:** Engineered a device-agnostic emergency pipeline on **AWS** (API Gateway, Lambda, DynamoDB, SNS to **Twilio** voice calls), provisioned with **Terraform** and tested end-to-end against **LocalStack**; wrote an asyncio TCP bridge with protocol adapters for Tuya WiFi buttons, 4G terminals, and GPS bracelets, reverse-engineering undocumented binary and text framing.
- **Aviators Spring:** Developed a **Flutter** pilot navigation app over a **Mapbox** base map, stacking FAA NASR airport data (a GeoJSON pipeline joining six source tables into simplified and full tiers), **NOAA** METAR/TAF weather, and 3.5 GB of VFR sectional charts published through an automated parallel tileset upload pipeline.

Oleyn — Dhaka, Bangladesh (Remote)
Founding Engineer (Backend & AI) — *May 2024 – Dec 2024*

- **Shrobon:** Engineered an AI-powered CRM and marketing campaign system featuring RAG chatbots. Developed a Bengali ASR system with fine-tuned models and **NVIDIA NeMo** speaker diarization for customer center analytics.
- **Superattorney:** Architected a legal research assistant using **FastAPI** and **Redis** caching for high-speed document parsing and retrieval.

Priyo Dhaka, Bangladesh (OnSite)

Software Engineer (Backend & AI) *Jun 2024 – Dec 2024*

- **Priyo-Penguin:** Deployed an AI customer support bot using **Django**, **Nginx**, and **OpenAI**, handling concurrent real-time queries.
- Developed **Dolphin-Priyo**, a warehousing application managing cross-platform users with **REST APIs**.
- Implemented bot protection using **Cloudflare Turnstile** and visualized data via **Metabase**.
- Managed dynamic email campaigns using **SendGrid** and Twilio integration.

Yobo Ontario, Canada (Remote)

Software Engineer Intern *Jan 2024 – May 2024*

- Built **YoboSQL**, a text-to-SQL AI chat interface using **LangChain**, **FastAPI**, and **SvelteKit**.
- Optimized database interactions with MariaDB and deployed on cPanel environments.

Key Projects →

DC-Ops *GRPO, OpenEnv, vLLM, Unsloth, PyTorch, ROCm*

- Built a physics-grounded reinforcement learning environment for datacenter operations using **OpenEnv**, featuring RC thermal networks and continuous multi-objective reward functions.
- Architected a highly efficient training pipeline on an **AMD MI300X**, generating expert reasoning trajectories with a 32B teacher via **vLLM** for initial Supervised Fine-Tuning (SFT).
- Designed a multi-signal GRPO reward system (physics simulation, scenario-aware heuristics, anti-looping, format compliance) that drove a **+188% composite reward improvement** and a **10× per-step gain** on hard multi-fault datacenter scenarios vs. the base model.

Scout *FastAPI, Langfuse, Docker, Terraform, Web-Scraping, Next.js, PostgreSQL, Agentic workflow*

- Engineered an autonomous competitive intelligence platform with scheduled agents to detect pricing changes and resolve them into a temporal knowledge graph.
- Architected an exactly-once job queue in **PostgreSQL** and orchestrated hosted browsers to safely render untrusted competitor content for automated weekly briefs.

AI Safety Forensics *Model Forensics, Preregistration, Cluster Bootstrap, llama.cpp, Qwen3.5-9B*

- Designed a model-forensics study on whether an agent that halts a concerning action after clarification was actually confused, or recovered through salience, scrutiny, or shifted priorities.
- Built a simulated file-selection task with misleading vs. accurate protection metadata; ran **Qwen3.5-9B** via **llama.cpp** through a pilot and a **preregistered** held-out confirmation, all 484 outcomes independently regraded (zero mismatches).
- Found that a factual user update cut protected-file selections to **2.8%** (1/36), vs. **75.0%** (27/36) after a neutral update, across 12 held-out task variants; an operator pre-review notice produced no comparable reduction.

Medical CoT Assistant *Unsloth, llama.cpp, Transformers, LoRA*

- Fine-tuned a 20B parameter LLM on medical datasets to enable clinical Chain-of-Thought (CoT) reasoning.
- Implemented an efficient training pipeline using **Unsloth** and **LLM-as-a-Judge** evaluation; optimized for deployment via **INT4 quantization** and GGUF export.

InsightAI *Spring Boot, FastAPI, Qdrant, LlamaIndex*

- Architected a microservices-based EdTech platform combining **Study Companion**, **Project Planner**, and **ShopGenie** modules.
- Developed **ShopGenie**, a multimodal e-commerce engine using **CLIP** embeddings for image search and **LLaVA** for product data augmentation.
- Built a RAG pipeline for the **Study Companion** that generates quizzes and flashcards from PDFs and YouTube videos using **LangChain** and **OpenAI**.
- Implemented a unified **Spring Boot** backend for data management and a **FastAPI** service for handling high-concurrency AI inference.

BD Prescription OCR *PyTorch, ResNet50, CRNN, SWA, MLflow, Grad-CAM*

- Designed a **ResNet50-CRNN** architecture that intercepts backbone feature maps and applies a **bidirectional LSTM** over the width axis, preserving left-to-right structure in handwritten word images for **78-class** medicine-name classification on a Bangladeshi prescription dataset.
- Built a robust training pipeline with **CLAHE** preprocessing, hash-based cross-split leakage detection, staged BatchNorm unfreezing, and **Stochastic Weight Averaging (SWA)**, progressively improving single-model accuracy from 86.08% to 89.82%.
- Ensembled 5 **StratifiedGroupKFold** SWA checkpoints (grouped by image hash to eliminate duplicate leakage), achieving a final test accuracy of **92.53%** and macro-F1 of **0.9234**, a **+6.44 pt** gain over the ResNet50 baseline.

Open Source Contributions, Benchmarks & Datasets

SREGym, live benchmark for AI SRE agents *2 merged PRs*

- Reported and fixed an **alert oracle** that rejected correct mitigations while the fault's own alerts were still clearing, so faster mitigations failed more often; it now waits for those alerts to resolve. [Issue #1116] [PR #1117]
- Added a **Kafka** poison-pill problem: one malformed record stalls a consumer while every pod still looks healthy. [PR #790]

GeoQL Environment, benchmark and RL environment for text to OverpassQL *Author*

- Built 72 tasks over 4 frozen **OpenStreetMap** worlds, each with 3 synthetic witness worlds where near-miss queries return different objects; a query passes only if it matches an accepted reading on every world. [Space] [Leaderboard]
- Evaluated 9 models in single-shot and **agentic** modes: witness worlds reject 28–30% of answers that are right on the real map, and a reference agent scores 100% on the real map alone but 0% on all worlds.

ITBench, IBM Research benchmark for IT automation *2 merged PRs*

- Reported problems in five **OpenTelemetry** demo scenarios and fixed three: a network policy fault that blocked nothing, an alert that fired on healthy idle services, and a fault aimed at a Deployment that does not exist. [Issue #1339] [PR #1340]
- Reported and fixed an **OpenTelemetry** demo install that timed out after a chart update renamed its span metrics connector. [Issue #1323] [PR #1324]

AIOpsLab, Microsoft framework for evaluating AIOps agents *1 PR in review*

- Reported that the **mitigation check** passes when the broken workload is deleted or scaled to zero, and opened a fix that also catches unschedulable pods. [Issue #208] [PR #209]

CloudOpsBench, benchmark of real-world cloud operations tasks for AI agents *2 PRs in review*

- Audited the v1.0 **graders** and grouped their defects by root cause: checks that pass when nothing was done, fail correct work, or crash. [Issue #14]
- Opened the first fixes: CI now runs every unit test, and the **Kinesis** policy grader checks that only the producer can write. [PR #15] [PR #16]

Kestra, workflow orchestration platform *2 merged PRs*

- Reported and fixed custom tool schemas that did not reach the **Anthropic API** intact. [Issue #147] [PR #148]
- Added a blueprint that classifies support tickets in **PostgreSQL** with **Claude Message Batches**. [PR #979]

Hugging Face Profile (@Melikshah) *LLM Fine-tuning, NLP Datasets, Multimodal Datasets*

- **GPT-OSS-20B-Clinical-CoT (GGUF):** Published a 4-bit quantized version of a 20B parameter model fine-tuned for clinical Chain-of-Thought reasoning, optimized for local inference.
- **Multimodal E-Commerce Datasets:** Scraped, curated and released the **Shajgoj** and **General Products** datasets, containing 20,000+ items for image-text retrieval tasks.
- **qwen3.5-4b-base-blindspots:** The Blindspot Discovery Dataset is a highly curated, adversarial evaluation set designed to stress-test and expose the architectural and logical limitations of the Qwen/Qwen3.5-4B-Base model.

Honors & Achievements →

Champion, IUT 11th National ICT Fest Hackathon (2024) – *1st Place in OpenAPI Category*
Champion, CodeCrafters Dev Sprint Hackathon, BUET (2024)
Champion, SUST CSE Carnival Hackathon (2024)
Runner-Up, Gen-Dev Hackathon by Acme AI (2024)
Champion, Cefalo ITverse Project Showcase (2023)
Honorable Mention, BCOLBD (Blockchain Olympiad) & IDSOL World Finalist (2024)
7th Place, SEC Inter University Junior Programming Contest (2022)

Technical Skills

Languages: Python, C++, Java, JavaScript, SQL
Frontend & Mobile: Next.js, React, SvelteKit, Flutter, Tailwind CSS
Backend: FastAPI, Django, Express, Firebase Cloud Functions, AWS Lambda, Spring Boot
AI & Machine Learning: PyTorch, Unsloth, ROCm, vLLM, Supervised Fine-tuning, Reinforcement Learning, llama.cpp, LangChain, LlamaIndex, RAG, Genkit, NVIDIA NeMo, HuggingFace
Databases & Vector DBs: PostgreSQL, MySQL, Redis, Firestore, ChromaDB, Qdrant, Supabase, Turso
DevOps & Tools: Docker, AWS (API Gateway, DynamoDB, SNS, SES), LocalStack, k8s, Terraform, GCP, Vercel, Nginx, Keycloak, PostHog, Stripe, Git

Certifications

Machine Learning Specialization | *Stanford Online & DeepLearning.AI*
Deep Learning Specialization | *DeepLearning.AI (Andrew Ng)*
PyTorch for Deep Learning Professional Certificate | *DeepLearning.AI*
Intermediate Problem Solver | *HackerRank*